

16. Develop a C program for implementing random access file for processing employee details.

AIM

To write a C program to implement **random access files** for storing and retrieving employee details using fseek().

CODE

```
#include <stdio.h>
```

```
struct Employee
```

```
{  
    int id;  
    char name[30];  
    float salary;  
};
```

```
int main()
```

```
{  
    FILE *fp;  
    struct Employee emp;  
    int n, pos;  
  
    fp = fopen("employee.dat", "wb+");  
  
    printf("Enter number of employees: ");  
    scanf("%d", &n);  
  
    for(int i = 0; i < n; i++)  
    {  
        printf("\nEnter Employee ID: ");  
        scanf("%d", &emp.id);
```

```
printf("Enter Employee Name: ");
scanf("%s", emp.name);

printf("Enter Salary: ");
scanf("%f", &emp.salary);

fwrite(&emp, sizeof(emp), 1, fp);
}

printf("\nEnter employee position to access: ");
scanf("%d", &pos);

fseek(fp, (pos - 1) * sizeof(emp), SEEK_SET);

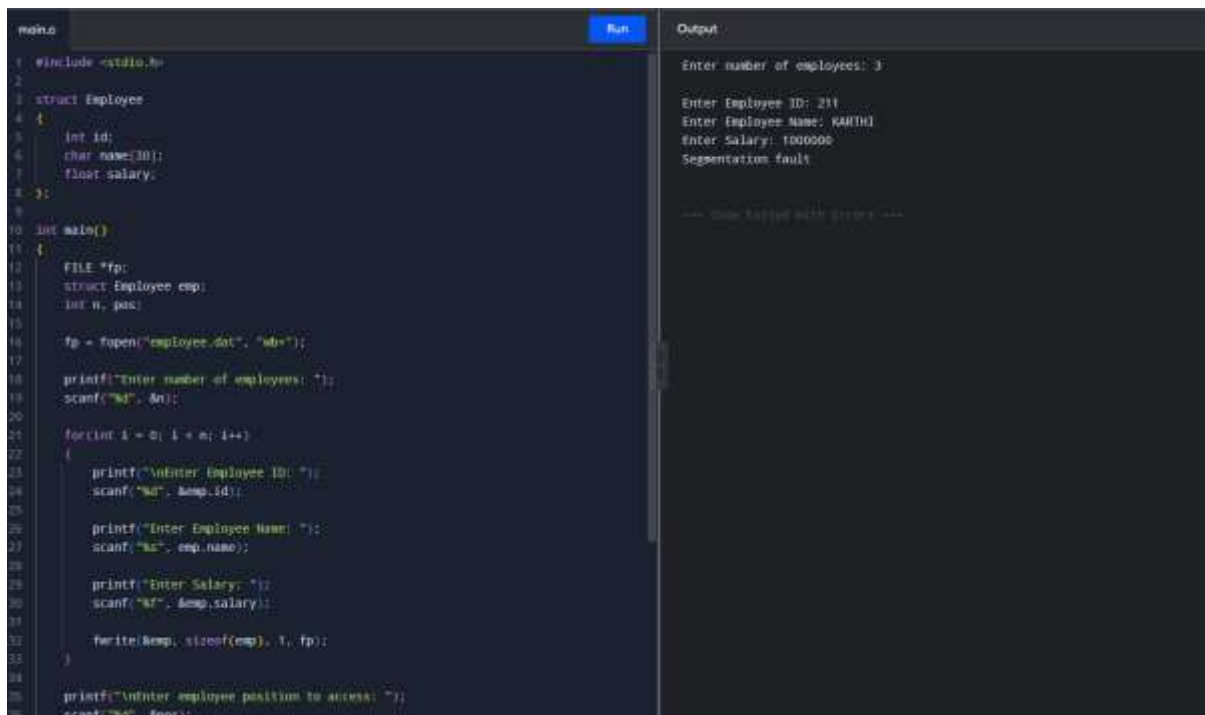
fread(&emp, sizeof(emp), 1, fp);

printf("\nEmployee Details\n");
printf("ID    : %d\n", emp.id);
printf("Name   : %s\n", emp.name);
printf("Salary : %.2f\n", emp.salary);

fclose(fp);

return 0;
}
```

OUTPUT:



The screenshot displays a C program in a code editor with a 'Run' button. The program defines an 'Employee' struct with fields for 'id', 'name', and 'salary'. It includes a 'main' function that prompts the user for the number of employees, then enters a loop to collect employee details (ID, Name, Salary) and writes them to a file named 'employee.dat'. The output window shows the user's input: 3 employees, with the first employee having ID 234, Name RAHUL, and Salary 1000000. A segmentation fault error is also visible in the output.

```
main.c Run Output
1 #include <stdio.h>
2
3 struct Employee
4 {
5     int id;
6     char name[30];
7     float salary;
8 };
9
10 int main()
11 {
12     FILE *fp;
13     struct Employee emp;
14     int n, pos;
15
16     fp = fopen("employee.dat", "wb+");
17
18     printf("Enter number of employees: ");
19     scanf("%d", &n);
20
21     for(int i = 0; i < n; i++)
22     {
23         printf("\nEnter Employee ID: ");
24         scanf("%d", &emp.id);
25
26         printf("Enter Employee Name: ");
27         scanf("%s", emp.name);
28
29         printf("Enter Salary: ");
30         scanf("%f", &emp.salary);
31
32         fwrite(&emp, sizeof(emp), 1, fp);
33     }
34
35     printf("\nEnter employee position to access: ");
36     scanf("%d", &pos);
```

Enter number of employees: 3
Enter Employee ID: 234
Enter Employee Name: RAHUL
Enter Salary: 1000000
Segmentation fault
*** Code exited with errors ***

RESULT

Thus, the C program for **random access file processing of employee details** was successfully implemented using `fseek()`, `fwrite()` and `fread()`.